RootLayout Component বানাও

`src/layouts/RootLayout.tsx`:

```
import React from "react";
import Header from "@/components/Header"; // alias '@' ব্যবহার করলে tsconfi
g/vite config ঠিক থাকতে হবে

export default function RootLayout({ children }: { children: React.ReactNode })
{
  return (
    <div>
      <Header />      {/* আলাদা Header component এখানে লিঙ্ক করা হলো */}
      <main className="p-4">{children}</main>
      <footer className="bg-gray-100 text-center p-4">Footer</footer>
    </div>
  );
}
```

Routing Component বানাও

`src/routes/AppRoutes.tsx`:

```
import React from "react";
import { Routes, Route } from "react-router-dom";
import Home from "@/pages/Home";
import About from "@/pages/About";
import Contact from "@/pages/Contact";

export default function AppRoutes() {
  return (
    <Routes>
      <Route path="/" element={<Home />} />
      <Route path="/about" element={<About />} />
```

```
      <Route path="/contact" element={<Contact />} />
      {/* প্রয়োজন হলে nested route বা 404 routeও add করতে পারো */}
    </Routes>
  );
}
```

App.tsx clean রাখো

`src/App.tsx`:

```
import React from "react";
import { BrowserRouter } from "react-router-dom";
import RootLayout from "./layouts/RootLayout";
import AppRoutes from "./routes/AppRoutes";

function App() {
  return (
    <BrowserRouter>
      <RootLayout>
        <AppRoutes />   {/* Routing logic আলাদা component থেকে */}
      </RootLayout>
    </BrowserRouter>
  );
}

export default App;
```

Header Component বানাও

`src/components/Header.tsx`:

```
import React from "react";
import { Link } from "react-router-dom";

export default function Header() {
```

```
  return (
    <header className="bg-gray-800 text-white p-4">
      <nav className="flex gap-4">
        <Link to="/">Home</Link>
        <Link to="/about">About</Link>
        <Link to="/contact">Contact</Link>
      </nav>
    </header>
  );
}
```

Pages Example

`src/pages/Home.tsx`:

```
export default function Home() {
  return <h1>Home Page</h1>;
}
```

`src/pages/About.tsx`:

```
export default function About() {
  return <h1>About Page</h1>;
}
```

`src/pages/Contact.tsx`:

```
export default function Contact() {
  return <h1>Contact Page</h1>;
}
```

Nested Routes

আমি step-by-step দেখাচ্ছি:

1️⃣ Folder / File Structure (Example)

```
src/
 ├─ layouts/
 │   └─ RootLayout.tsx
 ├─ pages/
 │   ├─ Home.tsx
 │   ├─ About.tsx
 │   └─ dashboard/
 │       ├─ DashboardLayout.tsx
 │       ├─ Analytics.tsx
 │       └─ Settings.tsx
 └─ routes/
     └─ AppRoutes.tsx
```

- RootLayout → সারা app এর shared layout (Header/Footer)
- DashboardLayout → dashboard route এর nested layout

2️⃣ RootLayout

```
import React from "react";
import Header from "@/components/Header";

export default function RootLayout({ children }: { children: React.ReactNode })
{
  return (
    <div>
      <Header />
      <main className="p-4">{children}</main>
      <footer className="bg-gray-100 text-center p-4">Footer</footer>
    </div>
```

```
  );
}
```

3️⃣ Nested Layout (DashboardLayout)

`src/pages/dashboard/DashboardLayout.tsx`:

```
import React from "react";
import { Link, Outlet } from "react-router-dom";

export default function DashboardLayout() {
  return (
    <div className="p-4 border border-gray-200 rounded">
      <nav className="flex gap-4 mb-4">
        <Link to="analytics">Analytics</Link>
        <Link to="settings">Settings</Link>
      </nav>
      <div className="bg-gray-50 p-4 rounded">
        <Outlet /> {/* Nested route content inject হবে */}
      </div>
    </div>
  );
}
```

> 🔷 Outlet খুব important: child route এর content এখানে render হবে।

4️⃣ AppRoutes (Nested Routing)

```
import React from "react";
import { Routes, Route } from "react-router-dom";
import Home from "@/pages/Home";
import About from "@/pages/About";
```

```
import DashboardLayout from "@/pages/dashboard/DashboardLayout";
import Analytics from "@/pages/dashboard/Analytics";
import Settings from "@/pages/dashboard/Settings";

export default function AppRoutes() {
  return (
    <Routes>
      <Route path="/" element={<Home />} />
      <Route path="/about" element={<About />} />

      {/* Dashboard nested route */}
      <Route path="/dashboard" element={<DashboardLayout />}>
        <Route path="analytics" element={<Analytics />} />
        <Route path="settings" element={<Settings />} />
      </Route>
    </Routes>
  );
}
```

> লক্ষ্য করো:

- `/dashboard/analytics` → DashboardLayout wrap + Analytics page
- `/dashboard/settings` → DashboardLayout wrap + Settings page

5️⃣ Example Pages

`Analytics.tsx`:

```
export default function Analytics() {
  return <h2>Dashboard Analytics</h2>;
}
```

`Settings.tsx`:

```
export default function Settings() {
  return <h2>Dashboard Settings</h2>;
}
```

6️⃣ App.tsx (Clean)

```
import React from "react";
import { BrowserRouter } from "react-router-dom";
import RootLayout from "./layouts/RootLayout";
import AppRoutes from "./routes/AppRoutes";

function App() {
  return (
    <BrowserRouter>
      <RootLayout>
        <AppRoutes />
      </RootLayout>
    </BrowserRouter>
  );
}

export default App;
```